6 The equation of the line L_1 is $y = 3x - 2$
The equation of the line L_2 is $3y - 9x + 5 = 0$

Show that these two lines are parallel.

L_1 has gradient 3.

$$3y - 9x + 5 = 0$$

(+9x −5) (+9x −5)

$$3y = 9x - 5$$

(÷3) (÷3)

$$y = 3x - \frac{5}{3}$$

1 mark
Rearrange L_2

1 mark
Equal gradients shown and compared

Both gradients are 3 and the intercepts differ, so the two lines are parallel.

(Total for Question 6 is 2 marks)